

Safety Analysis Report

Recent Stampedes in India and a Technology-Driven Crowd Safety Framework

Strategic Implementation Roadmap & AI Mitigation Strategies

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1. Recent Stampedes in India (2024–2025)

The following case studies highlight critical gaps in real-time crowd monitoring and flow regulation across various domains in India.

Hathras Crowd Crush (2024)

Type: Religious gathering | **Casualties:** ~121 Deaths

The congregation far exceeded capacity limits. The absence of segregated entry/exit routes and real-time density monitoring led to fatal compressive forces in narrow pathways.

Prayag Maha Kumbh Mela (2025)

Type: Mass Pilgrimage | **Casualties:** ~37 Deaths

Multiple converging streams caused barricade failure. Manual crowd control proved insufficient for ultra-high-density surges toward the Sangam area.

New Delhi Railway Station (2025)

Type: Transport Hub | **Casualties:** ~18 Deaths

Information shocks (delays) triggered overcrowding on a foot overbridge. Lack of unidirectional flow enforcement was a primary failure factor.

Karur Crowd Rush (2025)

Type: Political Rally | **Casualties:** ~41 Deaths

Attendance far exceeded permits. Sudden synchronized movement toward a focal attraction (the stage) caused a cascade of collapses.

2. Common Patterns & Failure Modes

- **Density Thresholds:** Rapid escalation beyond safe limits (e.g., 4-6 people/sqm).
- **Bottlenecks:** Failure at physical constraints like narrow gates or bridges.
- **Situational Awareness:** Delays in reporting surges from the field to command centers.
- **Information Shocks:** Sudden movements triggered by announcements or delays.

3. Proposed AI-Driven Framework

Our proposed system shifts from reactive response to predictive intervention using computer vision and temporal analysis.

3.1 Early Density & Flow Analysis

Using deep-learning models (e.g., CSRNet), the system generates real-time heatmaps to identify high-compression zones before they reach fatal levels.

3.2 Risk Scoring Logic

A composite score determines the threat level based on spatial and temporal variables:

$$\text{Risk Score} = \alpha(\text{Density}) + \beta(\text{Growth Rate}) + \gamma(\text{Flow Disorder}) + \delta(\text{Structural Constraint})$$

4. Alert & Response Hierarchy

Level	Description	Action Required
Normal	Density < 2 pax/sqm	Routine Monitoring
Monitor	Rising density / Bottlenecking	Deploy ground staff for flow control
High Risk	Density > 4 pax/sqm	Dynamic rerouting and entry restriction
Critical	Abnormal surge / Compression	Emergency evacuation & immediate intervention

5. Conclusion

Stampedes are rarely random. They follow detectable spatial and temporal patterns. By integrating vision-based density estimation with a robust alert logic, authorities can proactively manage mass gatherings and mitigate risks identified in the 2024-2025 tragedies.